

Arpita Chowdhury

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Education

Masters in Computer ScienceJan 2024 – May 2026

Texas State University, San Marcos, Texas, USA

- 3.67 out of 4.00 GPA

Bachelor of Science in Computer Science & EngineeringJan 2018 – Mar 2022

Chittagong Independent University, Chittagong, Bangladesh

- 3.40 out of 4.00 GPA

Work Experience

IT SupportAug 2024 – May 2026

Texas State University, San Marcos, Texas, USA

- Provided multi-shift systems support for 47+ real-time technical incidents spanning hardware access systems, software platforms, and digital service portals troubleshooting, resolving, and documenting each issue in a structured ticketing system while collaborating with IT teams, achieving a 97% first-contact resolution rate and measurably reducing system downtime across all supported operations.
- Maintained data integrity and audit-ready documentation across 135+ incident records by debugging and validating all ticket entries, troubleshooting steps, and resolution details ensuring 100% compliance with institutional data privacy policies and delivering consistent, reliable records that improved incident response time by 23%.
- Identified, prioritized, and coordinated resolution of 31 critical system issues across hardware, software, and network components within a 15-minute SLA implementing structured change control handoff procedures and version consistent escalation workflows that improved cross-team operational continuity by 31%.

Data AnalystMay 2022 - Dec 2023

Mega City Money Changers Ltd, Chittagong, Bangladesh

- Analyzed foreign currency exchange transactions across multiple currency pairs using Excel and SQL, identifying demand trends that improved pricing strategy accuracy by 22%.
- Audited and reconciled 2,800+ monthly transaction records by cross-validating source data against ledger entries, achieving 100% data accuracy and reducing manual reconciliation time by 35%.
- Flagged 14 anomalous transaction patterns through exploratory data analysis, supporting the finance team in maintaining regulatory compliance and reporting standards.

Technical Skills

- Cloud & Infrastructure: AWS (EC2, VPC, S3, IAM, CloudWatch, RDS, Lambda, DynamoDB, Route 53, CloudFront), GCP, Terraform, Pulumi.
- DevOps & Automation: Docker, Kubernetes, CI/CD (GitHub Actions, CodePipeline, CodeBuild, CodeDeploy).
- Data & Machine Learning: Pandas, NumPy, scikit-learn, XGBoost, TensorFlow, NLP, TF-IDF, Streamlit.
- Language and Developer Tools: Python, SQL, Bash, Git, Linux, SQLite, Power BI, Tableau.

Projects

IT Support Ticket Analytics Platform

- Built end-to-end data engineering and ML platform on 15,000 IT help desk tickets using Python, SQL, scikit-learn, and XGBoost. Engineered 34 features, trained ensemble classifier achieving 100% accuracy on 8-category ticket classification, deployed Isolation Forest anomaly detection (598 anomalies flagged), and delivered 8-page Streamlit dashboard with live ML inference, SLA breach monitoring, agent performance leaderboard, and volume forecasting. Automated full pipeline via GitHub Actions CI/CD with pytest test suite (24/24 passing) and AWS S3 + EC2 deployment.

Cloud Support Ticketing Simulation

- Simulated 3 production-level incident scenarios covering EC2 connectivity, CloudFront failover, and CI/CD pipeline failures using VPC Flow Logs, CloudWatch, and IAM Policy Simulator to diagnose root causes within 67 minutes and implement preventative measures, reducing recurring incidents by 31%.

Multi-Region High Availability Application

- Architected active multi-region infrastructure across us-east-1 and us west-2 using Route 53, multi-region RDS, and S3 cross-region replication with automated health checks, achieving 99.95% uptime and reducing failover time from 12 minutes to 47 seconds.

VPC Networking & Connectivity Troubleshooting (11 Labs)

- Diagnosed 27+ connectivity failures across public/private subnets, NAT gateways, and VPC peering using route table analysis, security group validation, and Flow Logs investigation, reducing average troubleshooting time from 2.1 hours to 41 minutes.

CI/CD Pipeline Implementation & Debugging

- Implemented end-to-end CI/CD pipeline using CodePipeline, CodeBuild, and CodeDeploy across 17 deployment runs, debugging 9 IAM permission failures with Policy Simulator and CloudWatch Logs analysis, improving deployment success rate by 33%.

Multi-Cloud Disaster Recovery (AWS + GCP)

- Developed cross-cloud disaster recovery solution using Pulumi for infrastructure-as-code with automated failover testing between AWS and GCP environments, conducting 7 failover drills that validated 100% recovery capability within 5-minute RTO.

Security Monitoring & IAM Troubleshooting (5 Labs)

- Implemented CloudWatch alarms, GuardDuty threat detection, and IAM Access Analyzer across 5 security monitoring scenarios, identifying 13 misconfigurations and resolving permission issues that improved security posture compliance by 29%.

Undergraduate Thesis — AI-Powered Bangla News Classification

- Developed ML model using TensorFlow and NLTK achieving 92% accuracy on 50,000+ articles scraped from 13 news portals; optimized training pipeline through data augmentation, reducing computation time by 40%.